

PHASE 2 – REQUIREMENT ANALYSIS

Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

College Name: Government Arts and Science College, Manur.

College Code: MSU142

Department: B.SC Computer Science

Team ID: SWTID-2026-9029

TeamLeader: Subbulakshmi M	NM ID: 7F7AC2ABD1D8C8DF6496517D478773
Team member: Arumuga Kani S	NM id: 01EE17CA3AAD3AB4EA4C8F841A16DD1
Team member: Mariammal C	NM id: BBE8324EB82C757A1054DE3C9951281A
Team member: Esakki Ganesh P	NM id: A705CB2C71F3AF38AF2473EFB5AE7725
Team member: Akalya R	NM id: B3F460E08A938D39C7F40868EA8EC1C6

1. Introduction

Requirement analysis is the process of identifying and documenting what the proposed system must accomplish. It defines the functional and non-functional requirements, users, inputs, outputs, and system constraints.

The objective of this project is to implement a Script-Controlled Access Control List (ACL) that restricts access to records based on the value of a selected field.

The system will dynamically evaluate the requested record and the current user's information before allowing or denying access.

2. Existing System

In a basic access-control system, permissions are commonly assigned according to user roles or groups.

For example:

- Administrator → Full access
- Manager → Limited management access

- Employee → Basic access

Although role-based permissions are useful, they may not be sufficient when access needs to be restricted at the individual record level.

Problems with the Existing Approach

- Users may have access to records they do not need.
- Role-based permissions can be too broad.
- Sensitive records may be exposed unnecessarily.
- Additional record-level restrictions may require more complex configuration.
- Static permissions may not consider the value stored in a particular record.

3. Proposed System

The proposed system introduces a script-controlled ACL that evaluates a field value before granting access to a record.

The ACL script can compare information associated with the current user against information stored in the requested record.

Example

Suppose each record contains a "Department" field.

If the user's department is IT:

User Department	Record Department	Result
IT	IT	Access Allowed
IT	HR	Access Denied
IT	Finance	Access Denied

This provides more granular control over record access.

4. Problem Requirements

The system should address the following requirements:

- Unauthorized users should not be able to access restricted records.
- Authorized users should be able to access records for which they satisfy the defined condition.
- The system should evaluate the relevant record field.
- The ACL should execute the required script during authorization.
- The access decision should be applied dynamically.
- The system should provide a clear way to test both allowed and denied access.

5. Functional Requirements

FR-01 – User Authentication

The system shall identify the user requesting access to a record.

FR-02 – Record Identification

The system shall identify the record requested by the user.

FR-03 – Field Value Retrieval

The system shall retrieve the value of the field used for access control.

FR-04 – User Information Retrieval

The system shall retrieve the relevant information associated with the current user.

FR-05 – Access Condition Evaluation

The system shall compare the required user information with the selected record field value.

FR-06 – Access Grant

The system shall allow access when the defined authorization condition is satisfied.

FR-07 – Access Denial

The system shall deny access when the authorization condition is not satisfied.

FR-08 – ACL Execution

The system shall execute the configured ACL script whenever the protected resource requires authorization.

FR-09 – Multiple Test Scenarios

The system shall support testing with both authorized and unauthorized users.

FR-10 – Security Enforcement

The system shall prevent users from bypassing the configured record-level access restriction through normal application access.

6. Non-Functional Requirements

NFR-01 – Security

The system should prevent unauthorized access to protected records.

NFR-02 – Performance

The ACL script should execute efficiently without causing unnecessary delays during record access.

NFR-03 – Reliability

The access-control mechanism should consistently apply the configured authorization condition.

NFR-04 – Maintainability

The ACL script should be structured clearly so that administrators or developers can modify the access condition when requirements change.

NFR-05 – Scalability

The approach should be capable of being extended to additional records, fields, users, and access conditions.

NFR-06 – Usability

Authorized users should be able to access permitted information without unnecessary additional steps.

7. User Requirements

Administrator

The administrator should be able to:

- Configure the required table and field.
- Configure the ACL.
- Create or manage test users.
- Define the access-control condition.
- Test the ACL behavior.
- Verify authorized and unauthorized access.

Authorized User

The authorized user should be able to:

- Log into the application.
- Access records that satisfy the defined condition.
- View permitted information.

Unauthorized User

The unauthorized user:

- May log into the application normally.
- Should not be able to access records that fail the ACL condition.
- Should receive an appropriate access-denied result.

8. Input Requirements

The system may use the following inputs:

Input	Description
User Identity	Identifies the current user
User Attribute	Department, role, location, or other relevant information
Record ID	Identifies the requested record
Protected Field	Field used to determine access
Field Value	Value stored in the protected record
ACL Condition	Rule used to make the access decision

9. Output Requirements

The system should produce one of the following outcomes:

Access Granted

The user satisfies the ACL condition and can access the requested record.

Access Denied

The user does not satisfy the ACL condition and access to the protected record is prevented.

10. Access-Control Logic

The basic authorization logic is:

Step 1: User requests a protected record.

Step 2: The ACL is triggered.

Step 3: The system identifies the current user.

Step 4: The system retrieves the relevant field value from the record.

Step 5: The script evaluates the authorization condition.

Step 6: If the condition is satisfied, access is granted.

Step 7: If the condition is not satisfied, access is denied.

Logical Representation

IF User_Attribute == Record_Field_Value

 THEN Allow Access

ELSE

 Deny Access

The exact comparison can be modified according to the project's selected use case.

11. Assumptions

The project assumes that:

- Users are properly authenticated.
- The required user information is available.

- The target records contain the field used for authorization.
- The ACL is correctly configured.
- The script has access to the information required for evaluation.
- Administrators have sufficient permissions to configure and test the ACL.

12. Constraints

The project may have the following constraints:

- Incorrect ACL configuration may cause unintended access behavior.
- Missing or incorrect field values can affect authorization.
- The implementation depends on the capabilities of the selected platform.
- The ACL script must be carefully designed to avoid security loopholes.
- Performance should be considered when authorization logic becomes complex.

13. Security Requirements

Security is the primary requirement of this project.

The system should:

- Verify the identity of the requesting user.
- Evaluate the required access condition.
- Prevent unauthorized record access.
- Avoid relying solely on client-side validation.
- Apply authorization on the server/platform access-control layer.
- Ensure that restricted records cannot be accessed through ordinary alternate access paths.
- Test both positive and negative authorization scenarios.

14. Requirement Traceability

Requirement ID	Requirement	Verification Method
FR-01	Identify current user	Functional Test
FR-02	Identify requested record	Functional Test

FR-03	Retrieve protected field value	Functional Test
FR-04	Retrieve user information	Functional Test
FR-05	Evaluate access condition	Functional Test
FR-06	Grant authorized access	Test Case
FR-07	Deny unauthorized access	Test Case
FR-08	Execute ACL script	ACL Verification
FR-09	Test multiple users/scenarios	Test Case
FR-10	Enforce record-level restriction	Security Test
NFR-01	Maintain security	Security Testing
NFR-02	Provide acceptable performance	Performance Observation
NFR-03	Provide reliable authorization	Repeated Testing
NFR-04	Maintainable implementation	Code Review
NFR-05	Support future extension	Design Review

15. Acceptance Criteria

The project will be considered successful when:

- The ACL is successfully configured.
- The required script executes during record access.
- A user satisfying the access condition can access the record.
- A user failing the access condition is denied access.
- The protected field is correctly evaluated.
- Multiple test scenarios produce the expected results.
- The final working process can be demonstrated through screenshots and a project demo.

16. Conclusion

The requirement analysis establishes the functional, non-functional, security, user, input, output, and testing requirements for the Script-Controlled ACL project.

The central requirement is to provide dynamic record-level access control based on a field value. The system should consistently grant access to authorized users and prevent access to records that do not satisfy the defined authorization condition.

This requirement analysis will serve as the foundation for the next phases: Project Design, Project Planning, and Project Development.